

# JACKSON MARSHALL

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## PROFILE

Second-year Chemical Engineering student at Dalhousie University seeking engineering co-op placement. Practical experience in large-scale operations management, inventory control, and technical process-following across high-accountability environments. Complementary skills in data tools, process documentation, safety compliance (WHMIS 2015), and cross-disciplinary software including Aspen Plus, SolidWorks, Python, and AutoCAD. Strong communicator; fluent in English and French.

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## EDUCATION

### Bachelor of Engineering, Chemical Engineering

Sep 2024 – Present (Expected 2029)

Dalhousie University · Halifax, NS

- Relevant coursework: Thermodynamics, Fluid Mechanics, Organic Chemistry, Applied Differential Equations, Engineering Design
- Active member of Dalhousie engineering community; pursuing co-op placement for Summer/Fall 2026

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## TECHNICAL SKILLS

**Engineering Software:** Aspen Plus, SolidWorks, AutoCAD, MATLAB (introductory)

**Programming & Data:** Python, JavaScript, HTML/CSS, Microsoft Excel (advanced), Google Workspace

**Engineering Concepts:** VLE flash calculations, psychrometrics, combustion analysis, Darcy-Weisbach fluid flow, recycle/purge systems, hydronic system design

**Safety & Compliance:** WHMIS 2015, First Aid & CPR, Bronze Cross, Google Cybersecurity Certificate

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## PROJECT & LAB EXPERIENCE

### Hydronic Heating System Design · ENGI 2103 — Fluid Mechanics

Winter 2026

- Designed a complete reverse-return hydronic heating system including pipe sizing, pressure drop analysis, and pump selection
- Applied Darcy-Weisbach equations and K-factor methods to model system resistance; referenced ASHRAE standards throughout
- Produced full engineering documentation including system schematics and design justification report

### Biodiesel Production Lab Report · CHEE 2203 — Organic Chemistry

Winter 2026

- Conducted and documented a transesterification experiment producing biodiesel from vegetable oil
- Analyzed product purity using chromatographic methods; reported yield, conversion, and process inefficiencies

### Aspirin Synthesis Lab Report · CHEE 2203 — Organic Chemistry

Winter 2026

- Synthesized aspirin via esterification reaction and performed recrystallization purification
- Reported theoretical vs. actual yield, potential error sources, and proposed process improvements

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## PROFESSIONAL EXPERIENCE

### Post-Production Assistant · Formosa Group

Toronto, ON | Apr 2025 – Jun 2025

- Managed secure logistics and movement of \$2M+ in sensitive production equipment; maintained chain-of-custody documentation
- Led systematic relabelling and inventory reconciliation of tens of thousands of individual assets
- Demonstrated the precision and procedural discipline directly transferable to regulated engineering environments

**Coordinator & Staff • Ashbridges Bay Volleyball***Toronto, ON | 2023 – Apr 2024*

- Coordinated scheduling and operations for a 100+ court facility using Google Calendar and Excel; oversaw \$10K+ in equipment logistics
- Enforced safety protocols across a large public facility with 250+ daily participants

**Bike Camp Instructor • Pedal Heads***Toronto, ON | Jun 2024 – Sep 2024*

- Supervised and instructed groups of youth in a daily outdoor program; managed safety, records, and parent communications

**CERTIFICATIONS & LANGUAGES**

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WHMIS 2015 • First Aid & CPR • Bronze Cross • CAN Sail 4 • Smart Serve • Google Cybersecurity Certificate

**Languages:** Fluent in English and French